

In 2016 E.c Model Exit Exam for Geology Department Students

1. Which of the following **sedimentary environments** is characterized by **sand, gravel and mud**?
 - a. active margin beach
 - b. glacial**
 - c. alluvial fans
 - d. deep marine
2. Which of the following is sinking of the ground naturally in almost **vertical direction**?
 - a. Subsidence**
 - b. Avalanche
 - c. Sliding
 - d. Flowage
3. Which of the following electrode configuration array is **only used** in the vertical electrical sounding (**VES**) surveying method?
 - a. Schlumberger array**
 - c. Dipole-Dipole array
 - b. Wenner-Schlumberger array
 - d. None of the above
4. During field work we have to record additional comment on the geological field map in case of.....
 - a. Degree of exposure is very low/absence
 - c. Area is covered by superficial cover
 - b. Exposure is covered by soil
 - d. All of the above**
5. Which one of the following is **correct** about **field maps**?
 - a. Using effective note book, representative photos and accurate structural measurement with the help of literature review field map can be prepared in the office
 - b. Perfectly recorded data on the note book can replace the field map we can prepare the map in the office with neatness.
 - c. During rush times a good geologist record all data on the field map so that he can prepare the field report based on the data recorded on the map alone.
 - d. Structural measurements and field data must be recorded both on the note book and on the field map no matter the conditions are for effective field work.**
6. Which one of the following is **true** about **isotropic minerals**?
 - a. Have different properties in different directions
 - b. Light travels through them in different ways with different velocities, depending on the direction of travel through a grain
 - c. Are always black when viewed under XP light
 - d. All**

7. Which one of the following is **true** about **opaque minerals**

- a. Do not transmit light in thin sections
- b. They appear black in both PP and XP light at all times
- c. Graphite and oxides are common opaque minerals
- d. All

8. Which one of the following is **true** about **gypsum**?

- a. Orthorhombic 2/m 2/m 2/m
- b. CaSO_4 Chemical composition
- c. Monoclinic 2/m
- d. Vitreous to pearly Luster

9. Which one of the following is a **phyllosilicates mineral**?

- a. Amphiboles
- b. Pyroxene
- c. Kaolinite
- d. Olivine

10. Which one of the following is **true** about **large scale Geological map**?

- a. Covers small area with greater detail information
- b. Covers wide area with less detail information
- c. Covers wide area with detail information
- d. Covers small area with less detail information

11. Which of the following are the folds with **parallel arrangement of limbs**?

- a. Reclined folds
- b. Inverted folds
- c. Isoclinal fold
- d. Symmetrical folds

12. The pressure and heat that **drive metamorphism result** from which **three forces**?

- a. The internal heat of the Earth, the weight of overlying rocks, and horizontal pressures developed as rocks become deformed
- b. Internal heat of the Earth, nuclear fission, heat released during chemical weathering
- c. Horizontal pressures developed as rocks deform, bonding, heat released during crystallization
- d. The weight of the overlying rocks, solar heating, and nuclear fusion

13. Which of following **basin is connected Melut basin of the Sudan** (currently Oil bearing basin)?

- a. Mekelle basin
- b. Gambella basin
- c. Ogaden basin
- d. Abay basin

14. Which one of the following is **not** a **type of groundwater recharge**?

- a. Hortonian overland flow **b. Direct recharge** c. Localized recharge d. Indirect recharge

15. Which of the following represents the **longest time period**?

- a. Mesozoic **b. Precambrian** c. Paleozoic d. Cenozoic

16. results from the mechanical lifting of air currents over natural barriers such as mountains.

- a. Orographic** b. Thunderstorms c. Convective d. Cyclonic

17. In a geochemical analysis of rocks, the best characterization for its source region will be from

- a. isotope studies** b. radiogenic isotopes c. non-radiogenic isotopes d. trace elements

18. Sr behaves like Ca element and the probable mineral hosting Sr isotopes are.....

- a. plagioclase** b. quartz c. apatite d. none of the above

19. **Most metallic minerals** of Ethiopia are associated with

- a. metamorphic rocks b. sedimentary rocks c. igneous rocks **d. all of the above**

20. During partial melting and fractional crystallization of magma,..... are concentrated in the liquid phase and become incorporated into the more silica- rich products.

- a. U and Th b. Hf and U c. Sm and Th **d. all of the above**

21. Which one of the following is **true** about **structural measurement**?

- a. Dip and plunge are measurements of inclination of linear features**
b. In recording structural measurement, strike and trend always represent in 3 digits whereas dip and plunge amounts are represented in 2 digits
c. In a data [087/46]; 087 refers Dip amount and 46 refers strike direction
d. Strike and trend are measured to know the horizontal continuity of planar features

22. Which one of the following **does not** include in **the scale of observation**?

- a. Regional observation c. Local observation
b. Exposure observation **d. Hand specimen observation**

23. Which of the following is **not** hydrothermal fluid?

- a. Connate water b. Meteoric water **c. Magmatic fluid** d. Hydrothermal solution

24. A combination of shallow earthquakes, tension, and contact metamorphism characterizes

- a. transform fault margins **c. spreading center**
b. subduction type margins d. continent/continent margins

25. Which one of the following are elements that commonly occur with sulfur in the Earth's crust?

- a. Atmophile elements b. Lithophile elements **c. Chalcophile elements** d. Siderophile elements

26. **Shear strength** of soil is a unique function of

- a. Effective stress only b. Total stress only
b. Both effective stress and total stress d. None of the above

27. Crustal contamination of rocks will lead to the enrichment ofisotopes.

- a. U-Th b. Sm-Nd **c. Rb-Sr** d. Sr-O

28. Which one of the following is **wrong** about the **components of Title page** in geological report writing?

- a. Contains date and place **b. Main purpose of the work**
b. Area name under investigation d. Contains name of researcher and investigator

29. **Paleoanthropology** is the study of

- a. Fossils of early reptiles and their descendants **b. Fossils of early humans and their ancestors**
c. Fossils of early birds and their ancestors d. Fossils of early fish and its descendants

30. Within a 50 mile traverse you walk from **shale** into a **slate** into a **phyllite**. You are walking in the direction of

- a. increasing degree of contact metamorphism b. Increasing in shock metamorphism
c. increasing metamorphic grade d. decreasing metamorphic grade

31. Which of the following is **correct** regarding the evaporation order of **evaporate minerals**?

- a. Calcite --dolomite -- gypsum-anhydrite (CaSO₄) -- halite (NaCl) -- K and Mg salts**
b. Dolomite ----- Calcite ----- gypsum-anhydrite (CaSO₄) ----- halite (NaCl) ----- K and Mg salts
c. K and Mg salts ----- dolomite Calcite gypsum-anhydrite (CaSO₄) -----halite (NaCl)
d. None of the above

32. Which one of the following is **correct** about **diamagnetic minerals**?

- a. The mineral responds a strong magnetic attraction
- b. The mineral have no response to a magnetic property**
- c. A material is drawn only weakly to an external magnetic field
- d. None

33. A set of folds with smaller wavelengths occurring superimposed on those with larger wavelength are described as

- a. Enechelon folds
- b. Disharmonic folds
- c. Digitations
- d. Parasitic folds**

34. During geological **report writing** the body of the report includes?

- a. Introduction, Literature review, Finding and discussion, Conclusion and recommendation
- b. Abstract, Introduction, Literature review, Finding and discussion, Conclusion and recommendation
- c. Abstract, Introduction, Literature review, Methodology, Finding, Discussion, Conclusion and recommendation, Reference**
- d. Introduction, Finding and discussion, Conclusion and recommendation

35. Which one of the following is geologic structure formed due to deposition of **sedimentary rocks**?

- a. soil erosion
- b. bedding**
- c. folding
- d. glacier erosion

36. Which one of the following is **wrong** about **dip direction**?

- a. It is the compass direction towards which the plane slopes
- b. It is the angle that the sloppy plane makes with the horizontal plane
- c. It's value varies from 0 to 360°** The value of dip direction typically varies from 0 to 180 degrees, not 360°
- d. It is orthogonal to strike

37. Commonly **incompatible elements** are of two groups i.e.....and.....

- a. high field strength and large ion lithophile
- b. large ion lithophile and trace elements**
- c. trace elements and REE
- d. none of the above

38. What is the **Law of Superposition**?

- a. All principle that states that rock layers were formed as horizontal layers
- b. A principle that states that the deeper a rock is, the older it is unless it has been disturbed or altered.
- c. A principle that states that an intrusive layer of igneous rock will always be younger than the rock it cuts through.
- d. A gap in rock layers due to erosion of a weak layer of rock

39. The maximum amount of dip of a bed is measured along **N40°E**. The strike direction of the bed is

- a. S50°W
- b. E-W
- c. N50°W
- d. N-S

40. Which of the following feature represent by **dot** on topographic map?

- a. Streams
- b. Mines
- c. Irrigation ditches
- d. Lakes

41. One of the following is **wrong** about **glassy texture**

- a. A completely glassy texture is called holohyaline texture
- b. Low rate of nucleation and growth of crystals
- c. Minerals are so fine, impossible to see without the help of microscope
- d. It is formed by very rapid cooling of magma

42. Mineral deposits that have formed simultaneously with the enclosing rock are called----

- a. Syngenetic
- b. Syngenetic
- c. Sinclogenetic
- d. Epigenetic

43. Which one of the following is the **"first melt"** produced by partial melting within the mantle, and which has not yet undergone any differentiation?

- a. Parental magma
- b. Igneous rock series
- c. Primary magma
- d. Ultramafic melt

44. Which one of the following geological map is suitable for **dam sit investigation**?

- a. Small scale geological map
- b. Detailed geological map
- c. Geological Reconnaissance map
- d. Specialized geological map

45. **Anions and cations** of isomorphic minerals have.....

- a. different coordination
- b. similarity of crystal structure
- c. different crystal structure
- d. different relative size

46. In general, **fine-grained** rocks have

- a. High permeability
- b. High porosity
- c. High porosity and low permeability
- d. High porosity and permeability

47. Which of the following statements hold **true** for **fossilization**?

- a. Only small animals are more likely to become fossils
- b. Fossilization is more likely to occur in mountains rather than in forests
- c. Every organism that dies becomes a fossil
- d. None of the above

48. Which of the following is **not ore mineral** of the **Greisen deposits**?

- a. Wolframite
- b. Lepidolite
- c. Cassiterite
- d. Chalcopyrite

49. Which is best classified as a **biochemical** sedimentary rock?

- a. Limestone that is made of calcite extracted from seawater by marine organisms
- b. Travertine that forms when calcium carbonate precipitates from water flowing through a cave
- c. Halite that forms when seawater evaporates
- d. Oolitic limestone that forms when calcite precipitates in spherical layers around a central fragment

50. **Electron microprobe analyses** used for the major elemental study of

- a. Minerals
- b. rocks
- c. rocks and minerals
- d. none of the above

51. Within the **lithosphere**, **water** is found in.....

- a. Solid state only
- b. Vaporous state only
- c. Liquid state only
- d. Liquid, solid and vaporous states

52. Which of the following is **not associated** with **stages in the life of mine**?

- a. Production rate
- b. Exploitation
- c. Exploration
- d. Development

53. The **Gulf of Aden fault system** controls

- a. Addis-Ambo-Nekemte lineament
- b. Adwa-Axum lineament
- c. Entoto escarpment
- d. All of the above

54. Gabbro and serpentinite are rich in magnetic minerals. If a geologist is standing beside a cliff which contains high amount of gabbro and serpentinite, which field equipment may show distortion/malfunction?

- a. Clinometer b. GPS c. Mobile phone/Camera d. Hand lens/binocular

55. Which one of the following statement is the **correct** statement?

- a. Quartz has high relief under a petrographic microscope
b. Crystallization is a process of change from a gas state to solid-state
c. The interference color of the mineral is the natural color of the minerals
d. Light-colored minerals are colorless in thin section

56. A rock rich in garnet and pyroxene that forms at extremely **high pressures** and **moderate to high temperatures** is called.....

- a. Amphibolite b. Hornfels c. Granulite d. Eclogite

57. During metamorphism, what is the **major effect of chemically active fluids**?

- a. increases the pressures in deeply buried, regional metamorphic zones
b. aid in the movement of dissolved silicate constituents and facilitate growth of the mineral grains
c. prevent partial melting so solid rocks can undergo very high temperature regional metamorphism
d. facilitate the formation of schistosity and gneissic banding in hornfels and slates

58. During geological **cross-section preparation** the following questions must be asked; **except?**

- a. Are the scale of the cross-section and the map are the same?
b. Is the line of the section marked on the map?
c. Does the line have kinks on it?
d. Does the cross-section fill the missing units on the map appearance?

59. Which of the following elements are **large** ion lithophile elements?

- a. K, Rb & Ba b. K, Rb & O c. H, N & O d. Sr, Rb & S

60. **Stable isotopes** are frequently out of equilibrium in rocks due to

- a. Magma chamber processes
- b. **Equilibration with a fluid phase following crystallization**
- c. High temperatures
- d. Low temperature and pressure

61. Problems of foreign companies in Ethiopia are.....

- a. extreme limitation of financial and expert capacities
- b. false exploration report preparation
- c. hiring very expensive foregone experts
- d. **All of the above**

62. Which one of the following is **not likely** to be genetically associated with impact of an asteroid or large meteorite?

- a. Diamond inclusion
- b. **Blueschists**
- c. Impact crater
- d. Coesite

63. If you take a carbon date on a mammoth bone in a Pleistocene deposit, what type of dating and stratigraphy method is this for the Pleistocene deposit?

- a. Biostratigraphy, relative dating
- b. Chronostratigraphy, relative dating
- c. Lithostratigraphy, relative dating
- d. **Chronostratigraphy, absolute dating**

64. Which of the following is **different** from the others?

- a. Geophysical fingerprint
- b. **Geophysical ambiguity**
- c. Geophysical anomaly
- d. Geophysical signature

65. If the mineral contains no planes of weakness, it will **break** along random directions this physical propriety is called

- a. Parting
- b. Hardness
- c. Cleavage
- d. **Fracture**

66. In which of the following range projected coordinate system or metric system is effective and beyond that distortion will be high and measurement will be inaccurate?

- a. **800N840S**
- b. 750N750S
- c. 800N - 800S
- d. 840N-800S

67. Which one of the following activity does **not include** among main geological field activity domains?

- a. Geological map preparation and analysis
- b. Observing and recording field data selectively and systematically
- c. Making careful observations of rock types, textures, structures
- d. Describing the lateral and vertical distribution of litho-facies of stratigraphic units

68. Where is the origin of Earth's magnetic field generated?

- a. Earth's Mantle
- b. Earth Inner core
- c. Earth Outer core
- d. Earth crust

69. The following are the surficial processes that form mineral deposits **except one**:

- a. Weathering enrichment and mechanical concentration
- b. Low-temperature precipitation
- c. Formation by weathering
- d. Magma

70. How many major **river basins** are there in Ethiopia?

- a. 13
- b. 10
- c. 11
- d. 12

71. Which of the following is **wrong** about **Universal Transversal Mercator** (UTM)?

- a. It is type of cylindrical projection
- b. There are 90 longitudinal zone (Lune) to cover the whole globe
- c. It subdivided the globe in to portions called longitudinal zone (Lune)
- d. It begin at 1800W and working eastwards

72. Which of the following is **true** about **magma differentiation**?

- a. Degree of crystal growth increase as temperature increase
- b. Amount of Ca, Mg and Fe increases as temperature increases
- c. Amount of silica increase as temperature increases
- d. Amount of K, Na and Al increases as temperature decreases

73. The **highest** degree of symmetry is shown by..... and the **lowest** degree of symmetry is shown by_ respectively:

- a. Cubic system and monoclinic system
- b. Hexagonal system and monoclinic system
- c. Tetragonal system and monoclinic system
- d. Cubic system and triclinic system

74. **Precambrian** geology of Ethiopia is **mainly focused** on the study of

- a. **Metamorphic rocks** b. Volcanic rocks c. Extrusive rocks d. Sedimentary rocks

75. The following activities are phase of geological field work, **except one?**

- a. Post field activities b. Lab-Analysis c. Syn-Field work **d. Office work**

76. The centre of the earth (inner core) is very hot, thick and despite the high temperature it is solid. What is the **main reason** that prevents melting?

- a. Absence of fluid which activate melting
b. Intense pressure which exceed the temperature
c. The presence of very dense elements with higher melting point than the core temperature
d. All of the above is true

77. Which of the following is **false** about **pyroclastic flow/surge?**

- a. It can move with great force and speed, hence can destroy small building
b. Surge can burn and bury anything on its path
c. It is the most disastrous volcanic event following basic magma eruption
d. It is the combination of hot ash and poisonous gas

78. Elements that **do not** readily fit into the minerals structure are call.....

- a. compatible element **b. incompatible elements** c. major element d. trace element

79. Which of the following geophysical methods of exploration is of particular significance in defining oilfield traps such as anticlines and salt domes?

- a. Geomagnetic method b. Seismic refraction c. Electromagnetic method **d. Gravity method**

80. The following land forms are an example of basic lava flows; **except?**

- a. Pillow lava b. Pahoehoe Lava c. Aa lava **d. Lava domes**

81. Which two rocks are composed primarily of **quartz, feldspar, and clay minerals?**

- a. sandstone and limestone c. rock salt and conglomerate
b. rock salt and breccias **d. sandstone and shale**

82. When the rainfall is unevenly distributed over the area, for instance due to orographic effects, the method may give good results.

- a. Arithmetic average** b. Isohyetal c. Statistical d. Thiessen polygon

83. Which of the following methods is most accurate for the determination of the water content of soil

- a. Oven drying method
- b. Calcium carbide method
- c. Sand bath method
- d. Pycnometer method

84. During dam site investigation what is the shape of valley usually preferred for a reservoir?

- a. V-shaped only
- b. Flat lands or plains
- c. U or V-shaped
- d. U-shaped only

85. Which of the following is wrong regarding the cutoff grade?

- a. It is useful at the operational level in distinguish ore from the waste
- b. It is the minimum min-able grade
- c. It is the valuable solid mineral that is sought and extracted from the working of a mine
- d. It is determined by the operating cost and price of minerals in the world market

86. Continuous lines plotted on topographic map following parallel/subparallel path and never cross each other is called.....?

- a. Stream lines
- b. Contour lines
- c. Traverse lines
- d. Main roads

87. Which of the following statement is false?

- a. Actual evapotranspiration is the evaporation from a vegetal cover under the natural or given condition of supply of moisture
- b. Evapotranspiration is anthropogenic process where water vapour is transferred from both land surface and vegetation surface by evaporation and transpiration processes.
- c. Potential evapotranspiration is the evapotranspiration from a vegetal cover if sufficient water is supplied to obtain optimum growth or the maximum amount of vapor which might be transferred under the existing meteorological conditions (water is not the limiting factor)
- d. Evaporation is the transfer of water into the atmosphere from a free water surface, a bare soil or interception into a vegetal cover.

88. In which of the following sedimentary environments would you least expect to find gravel?

- a. alluvial fans
- b. deep marine
- c. active margin beach
- d. glacial

89. Which of the following rocks could be an important source of diamonds?

- a. Lamprolite
- b. Granite
- c. Kimberlite
- d. A and C

90. Which one of the following is **well-developed crystal** with most crystal faces shown?

- a. Dendritic crystals b. Subhedral crystals c. Anhedral crystals **d. Euhedral crystals**

91. In recording paleontological data where should we start looking for fossils?

- a. Start in hard and intact rock unit where there is loose pieces of rocks are there
- b. Start in environment where sedimentation rate is low, and unfavorable for rapid mineralization
- c. Start in reducing environment, fine-grained sedimentary deposits such as mudstones**
- d. Start in red colored, coarse grained sedimentary deposit where oxygen is abundant

92. What comes from trees and preserves insects and other small animals?

- a. Amber** b. Tar c. Carbon d. Ice

93. Which of the following statement **true** about **triaxial stress system**?

- a. The three principle stresses are equal
- b. The three principle stresses are unequal
- c. The three principle stresses are unequal and have non-zero values**
- d. None of the above

94. Which one of the following events occurs in the formation of **sedimentary rock**?

- a. Weathering** b. Melting c. Cooling d. Crystallization

95. Geological data **may not be** collected in the form of.....

- a. Photography b. Measurement
- b. Questioners to the local people** d. Notes and physical samples

96. The discriminant diagram of trace elements must be prepared by taking into consideration the elements which are

- a. insensitive to secondary processes b. easy to measure even at low concentration
- c. relatively simple and rapid method of analyses **d. all of the above**

97. **Over thrusts** can be describes as

- a. Low angle thrust faults b. High angle reverse faults
- c. High angle reverse faults with large net slip **d. Low angle thrust faults with large net slip**

98. Which of the following is false?

- a. As grain size of rocks increases, the abrasion resistance of rocks (Strength of rocks) increases.
- b. As temperature increases elastic modulus and compressive decreases.
- c. Strength increases as the aspect ratio increases
- d. As confining pressure increases, intact rock strength increases.

99. Which of the following inferring techniques is advisable for the part of the study is where traversing could not cover?

- a. By looking topographic features
- b. By following contour lines
- c. By painting common color for the uncertain
- d. A and B are the answers

100. Which of the following method is suitable for groundwater exploration?

- a. Electrical method
- b. Gravity method
- c. Magnetic method
- d. Seismic method